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Multi-container filling machine technologies

Abstract

An apparatus for closing fluid containers includes a rotary gassing station. In the rotary gassing station, at least one container containing a quantity of fluid within an interior space thereof receives a quantity of gas in the interior space. A rotary seaming station is positioned proximate to the rotary gassing station. In the rotary seaming station, a lid is seamed to an opening of the container to seal the quantity of fluid and the quantity of gas in the interior space. At least one container piston of the rotary seaming station is movable around a center axis of the rotary seaming station. The container piston holds the container between a first radial position when the container is within the rotary gassing station to at least a second radial position when a lid is seamed to an opening of the container in the rotary seaming station.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part application of U.S. application Ser. No. 16/831,458 filed Mar. 26, 2020, which claims benefit of U.S. Provisional Application Ser. No. 62/824,862, entitled, "Multi-Container Filling Machine Technologies" filed Mar. 27, 2019, the entire disclosures of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure is generally related to container filling machines and more particularly is related to multi-container filling machine technologies.

BACKGROUND OF THE DISCLOSURE

(2) A variety of types of filling machines are used throughout the food and beverage industries to fill containers with beverages and liquid food products. Many large productions utilize filling machines that are designed to fill a specific container type, which has a specific container dimension and fluid volume. These machines are commonly expensive and only used by large-scale productions. Small productions, such as micro-breweries, are often unable to afford these large-scale machines due to their high cost and the large-scale production of goods that makes them economically viable. As a result, small productions must resort to having their products packaged off-site by third party companies, or utilize packages or containers which are different from what the production company desires.

(3) In addition to these above-noted shortcomings in the industry, there are a number of other drawbacks that come with using conventional filling machines to which the subject disclosure provides substantial improvements over. These drawbacks of the conventional filling machines may include issues relating to the efficiency of operation and the mechanical and electrical components used with the machines.

(4) One particular issue relates to process efficiency of the filling machine and how, typically, each stage of the filling process is handled by a separate unit or piece of machinery which is connected by a belt or another device for transporting containers between the separate unit. As an example, in conventional filling machines, after a container is filled with a beverage at a filler station, the container moves along a transportation belt for a particular distance, usually 10-15 feet, until it reaches a seaming station, where the container is gassed and seamed. Having separate machines handling different parts of the process is beneficial, especially with regards to separate manufacturing and installation of the machines, and the use of separate machinery is carried over from the origin of beverage container manufacturing where the manufacturer of the container would often provide a device of lidding or sealing that container, while the filler machine to fill the container with a beverage is manufactured by a separate company. However, the use of separate

machines separated by transportation belts in modern settings is often not ideal, since it occupies a greater spatial footprint and requires installation of numerous container transportation devices.

(5) Thus, a heretofore unaddressed need exists in the industry to address the aforementioned deficiencies and inadequacies.

SUMMARY OF THE DISCLOSURE

(6) Embodiments of the present disclosure provide an apparatus for closing fluid containers. Briefly described, in architecture, one embodiment of the apparatus, among others, can be implemented as follows. The apparatus has a rotary gassing station. In the rotary gassing station, at least one container containing a quantity of fluid within an interior space thereof receives a quantity of gas in the interior space. A rotary seaming station is positioned proximate to the rotary gassing station. In the rotary seaming station, a lid is seamed to an opening of the at least one container to seal the quantity of fluid and the quantity of gas in the interior space. At least one container piston of the rotary seaming station is movable around a center axis of the rotary seaming station. The at least one container piston holds the at least one container between a first radial position when the at least one container is within the rotary gassing station to at least a second radial position when the lid is seamed to the opening of the at least one container.

(7) Embodiments of the present disclosure provide an apparatus for closing fluid containers. Briefly described, in architecture, one embodiment of the apparatus, among others, can be implemented as follows. The apparatus has a container infeed. A filling station receives at least one container from the container infeed. In the filling station, the at least one container is at least partially filled with a quantity of fluid in an interior space of the at least one container. A rotary gassing station receives the at least one container from the filling station. In the rotary gassing station, the at least one container containing the quantity of fluid within the interior space thereof receives a quantity of gas in the interior space. A rotary seaming station is positioned proximate to the rotary gassing station. In the rotary seaming station, a lid is seamed to an opening of the at least one container to seal the quantity of fluid and the quantity of gas in the interior space. At least one container piston of the rotary seaming station is movable fully around a center axis of the rotary seaming station. The at least one container piston holds the at least one container between at least first radial position when the at least one container is within the rotary gassing station to at least a second radial position when the lid is seamed to the opening of the at least one container.

(8) Embodiments of the present disclosure provide an apparatus for closing fluid containers. Briefly described, in architecture, one embodiment of the apparatus, among others, can be implemented as follows. The apparatus has a rotary seaming station. At least one container lift piston is in the rotary seaming station. The at least one container lift piston holds at least one container at and between a first radial position when the at least one container is within a rotary gassing station to at least a second radial position when a lid is seamed to an opening of the at least one container in the rotary seaming station.

(9) Other systems, methods, features, and advantages of the present disclosure will be or become apparent to one with skill in the art upon examination of the following drawings and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the present disclosure, and be protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings,

like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1 is a side view illustration of an inline container filling machine, in accordance with a first exemplary embodiment of the present disclosure.

(3) FIG. 2 is a side, isometric view illustration of the inline container filling machine of FIG. 1 illustrating further detail, in accordance with the first exemplary embodiment of the present disclosure.

(4) FIGS. 3A-3B are side, isometric view illustrations of a narrow belt and lifting device of the inline container filling machine of FIG. 1, in accordance with the first exemplary embodiment of the present disclosure.

(5) FIGS. 4A-4C are front view and side, partial cross-sectional view illustrations, respectively, of a narrow belt and lifting device of the inline container filling machine of FIG. 1, in accordance with the first exemplary embodiment of the present disclosure.

(6) FIGS. 5A-5B are front, partial cross-sectional and side, partial cross sectional view illustrations, respectively, of a narrow belt and lifting device of the inline container filling machine of FIG. 1, in accordance with the first exemplary embodiment of the present disclosure.

(7) FIG. 6 is a side view illustration of a rotary container filling machine, in accordance with the first exemplary embodiment of the present disclosure.

(8) FIG. 7 is an elevated, front isometric view illustration of the rotary container filling machine of FIG. 6, in accordance with the first exemplary embodiment of the present disclosure.

(9) FIG. 8 is an elevated, side isometric view illustration of the rotary container filling machine of FIG. 6, in accordance with the first exemplary embodiment of the present disclosure.

(10) FIGS. 9-10 are plan view illustrations of a container filling machine, in accordance with a second exemplary embodiment of the present disclosure.

(11) FIGS. 11-22 are illustrations of components and operation of the rotary gassing station of the container filling machine of FIGS. 9-10, in accordance with the second exemplary embodiment of the present disclosure.

(12) FIG. 23A is a top view illustration of an apparatus for closing fluid containers, in accordance with a third exemplary embodiment of the present disclosure.

(13) FIG. 23B is a top view diagram of the apparatus for closing fluid containers of FIG. 23A, in accordance with the third exemplary embodiment of the present disclosure.

(14) FIG. 24 is a plan view illustration of a rotary seaming station of the apparatus for closing fluid containers, in accordance with the third exemplary embodiment of the present disclosure.

(15) FIG. 25 is a top view illustration of the rotary seaming station of FIG. 24, in accordance with the third exemplary embodiment of the present disclosure.

(16) FIGS. 26A-26C are plan view illustrations of the rotary seaming station of the apparatus for closing fluid containers, in accordance with the third exemplary embodiment of the present disclosure.

(17) FIGS. 27A-27F are top view illustrations of the rotary seaming station of the apparatus for closing fluid containers showing a sequential process, in accordance with the third exemplary embodiment of the present disclosure.

(18) FIGS. 28A-28B are plan view illustrations of the rotary seaming station of the apparatus for closing fluid containers, in accordance with the third exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

(19) FIG. 1 is a side view illustration of an inline container filling machine 10, in accordance with a first exemplary embodiment of the present disclosure. As shown in FIG. 1, the container filling machine 10, which may be referred to simply as 'machine 10', is an inline filling machine that is designed to move containers 12 along a substantially linear path through a filling operation. In contrast to inline filling machines, the subject disclosure may also be applicable to rotary filling machines, as discussed relative to FIGS. 6-8.

(20) The operation of filling the containers **12** may begin with appropriate cleaning of the containers **12** and loading them on to the machine **10**. The containers **12** may be moved on a belt **20** which is positioned underneath the containers **12** such that the bottom surface of the containers **12** contacts a top surface of the belt **20**. The belt **20**, which may also be referred to as a conveyor, may be formed from a number of linkages connected together, e.g., metal, plastic, or other linkages, or from one continuous belting material, such as a rubberized belt. On the sides of the belt path, various guardrails **22** may be used to ensure the containers remain on the belt **20**.

(21) A predetermined number of containers **12**, such as six (6) containers as shown in FIG. **1**, may be moved via the belt **20** to a position below a counter pressure filling tank **30** which contains a quantity of filling liquid (or other filling substance) which is dispensed into the containers **12** through a plurality of filling heads **32**. Appropriate nozzles, valves, and other electro-mechanical equipment may be used to complete the filling of the containers **12**, including, for example, filling lifting devices **34** which raise a container **12** off the belt **20** so an exposed opening of the container **12** can be engaged with a filling nozzle. To maintain the containers **12** in the appropriate position under the filling heads **32**, one or more stop gates **40** may be used to physically prevent movement of the containers **12** as the belt **20**, on which the containers **12** are positioned, maintains a constant movement. Alternatively, the belt **20** may be designed to stop and start movement to position the containers **12** in the desired location.

(22) After the containers **12** are filled, the belt **20** transports them to a staging area **42** where the containers **12** are waiting to be closed with lidding and sealing operations. Another group of containers **12** may then be moved to the filling station to provide a continuous operation of the device **10**. At a lidding station **50**, a lidding device **52** may dispense individual lids **54** on to the containers **12** as they pass underneath the chute of the lidding device **52**. For example, as the container **12** is moved via the belt **20** past the lidding device **52**, an individual lid **54** may be held above the container **12** such that the container **12** makes contact with the lid **54** as it moves past. The contact between the container **12** and the lid **54** causes the lid **54** to fall on to the top opening of the container **12**. After the lidding station **50**, a stop gate **40** may prevent the container **12** from moving to the seaming station **60** until the desired time.

(23) At the seaming station **60**, a seaming device **62** may be used to seal the lid **54** on to the top opening of the container **12**, thereby sealing the beverage or other substance contained within the container **12** inside it. While this disclosure discusses a seaming station **60**, it is noted that the station is generally understood as a closure station, whereby the actual sealing of the containers **12** may be achieved with capping, crowning, sealing, or another process. FIG. **1** depicts two seaming devices **62** used on the filling line, such that two containers **12** can be sealed concurrently. In this regard, when the containers **12** go past the lidding station **50**, a series of gates **40** can be used to stop the containers **12** at one of two (or more) seaming devices **62** which can then seal the containers **12**. The use of two or more seaming devices **62** can increase production of the machine **10** because the containers **12** can be sealed twice as fast (with two seaming devices **62**—or even faster with more seaming devices). It is estimated that in full production, the machine **10** with two seaming devices **62** may be capable of achieving a 100 container/minute production rate, which is significantly higher than comparably-sized machines. The seaming devices **62** may use a closure lifting device **64** which raises the container **12** off the belt **20** so the top opening of the container **12** can have the lid sealed thereto. Once the containers **12** are sealed, they may be moved to a subsequent step in the production process, such as to cleaning and packaging.

(24) It is noted that the production line of the machine **10** may include many additional components, features, and functions which are not explicitly detailed herein. For example, a CO₂ atmosphere may be applied to all or part of the operation line, thereby preventing the beverages from being exposed to oxygen during production. It is also noted that the containers **12** may be any type of container which is used to hold a substance, commonly a beverage or food, but also non-edible substances. The containers **12** may have varying sizes and shapes, and they may be

constructed from different materials. For example, the containers may be aluminum cans, glass or plastic bottles, growlers, champagne bottles or the like. In accordance with this disclosure, the machine **10** is described relative to filling canned beverage containers, such as those commonly used to contain soda or beer, but other containers may also be used with the machine **10**.

(25) FIG. **2** is a side, isometric view illustration of the inline container filling machine **10** of FIG. **1** illustrating further detail, in accordance with the first exemplary embodiment of the present disclosure. In particular, FIG. **2** illustrates additional details of the lidding and seaming stations **50**, **60**. With reference to FIGS. **1-2**, as can be seen, the containers **12** are showing receiving the lids **54** from the chute of the lidding device **52**. The lids **54** may be stacked within the lidding device **52** and dispensed on to the container **12** openings as they pass underneath. The seaming station **60** may be positioned immediately adjacent to the lidding station **50**, such that the containers **12** can be appropriately sealed as quickly as possible. It is noted that FIG. **2** illustrates only a single seaming station **60**, but additional seaming stations can be included.

(26) FIGS. **3A-3B** are side, isometric view illustrations of a narrow belt **20** and lifting device **64** of the inline container filling machine **10** of FIGS. **1-2**, in accordance with the first exemplary embodiment of the present disclosure. With reference to FIGS. **1-3B**, and as can be seen in FIGS. **3A-3B** specifically, the belt **20** is a narrow belt formed from a plurality of linkages or interconnected members which together are capable of conveying the containers **12** through the machine **10**. This type of belt **20** may be generally referred to as a 'narrow belt' because the width of the belt **20** is less than the diameter distance of the container **12**. It is noted that the term 'narrow belt' may be used relative to the container size with which the belt is used, such that the actual width of the narrow belt **20** can vary but will always remain less than the diameter of the container on the belt.

(27) In contrast to the use of a narrow belt **12**, conventional devices commonly have a belt which is wider than the container's diameter, such that the belt can properly transport the container. However, with these conventional devices, when the container needs to be raised or lifted off the belt, such as during filling or seaming, the container must be pushed off the belt to allow for a lifting device to lift the container **12** without contacting the underside of the belt. For example, most conventional fillers need to either have a star wheel to move the containers off of the belt to add the lids or caps, or they push the containers off a belt to add caps, crowns, then be pushed to another belt to move them away. The use of the narrow belt **20** can avoid the need to move the containers **12** off of the belt **20** for these processes.

(28) In particular, the narrow belt **20** allows for a lifting device **64** to raise a closure lift mechanism **66** on either side of the narrow belt **20**, such that the lifting device **64** can raise the container **12** off of the narrow belt **20** vertically. The closure lift mechanism **66** may include two or more members which are positioned on the lateral sides of the narrow belt **20** and which each have a rotatable guide wheel **68**. The lifting device **64** may raise and lower the closure lift mechanism **66** and the rotatable guide wheels **68** relative to the top surface of the narrow belt **20**. For example, in a retracted position (non-lifting position) of the lifting device **64**, the uppermost part of the closure lift mechanism **66** and the rotatable guide wheels **68** may be positioned below the top surface of the narrow belt **20**, such that containers **12** can freely move along the narrow belt **20** without contacting the closure lift mechanism **66** and the rotatable guide wheels **68**. In a raised or lifted position, the closure lift mechanism **66** and the rotatable guide wheels **68** may raise to a position above the upper surface of the narrow belt **20**, as shown in FIG. **3B**. The operation of the machine **10** may be controlled with appropriate sensors to raise the closure lift mechanism **66** and the rotatable guide wheels **68** when a container **12** is positioned immediately vertical or overhead, such that the rotatable guide wheels **68** make contact with the underside or bottom of the container **12** and raise it upwards. In the raised position, the container **12** may be sealed or capped appropriately, at which point the lifting device **64** can lower the closure lift mechanism **66** and the rotatable guide wheels **68** back downwards until the container **12** is sitting on the narrow belt **20** again.

(29) As can be seen, with the narrow belt **20** design, the containers need only be simply stopped with a stop gate **40** while they are still on the belt. Then the lifting device **64** from underneath, and from both sides of the narrow belt **20**, raises the containers **12** to add a cap, crown or spin the can if a lid was placed on top beforehand. The use of the narrow belt **20** is important for allowing the closure lift mechanisms **66** and the rotatable guide wheels **68**—or other equivalent structures—to contact the bottom of the container **12** without needing to contact the belt **20** itself.

(30) FIGS. **4A-4C** are front view and side, partial cross-sectional view illustrations, respectively, of a narrow belt **20** and lifting device **64** of the inline container filling machine **10** of FIG. **1**, in accordance with the first exemplary embodiment of the present disclosure. In particular, FIG. **4A** depicts the container **12** positioned on the narrow belt **20** with the lifting device **64** in the lowered or retracted position. As can be seen here, the guide wheels **68** are connected to the closure lift mechanisms **66** which are raised and lowered by the lifting device **64**, and in the lowered or retracted position, the bottom **14** of the container **12** can move along the narrow belt **20** without contacting the guide wheels **68**. FIG. **4B** depicts the lifting device **64** in the raised or extended position, where the guide wheels **68** are moved upwards on either side of the narrow belt **20** to contact the bottom **14** of the container **12** and raise it vertically off the narrow belt **20**. As can be seen, the narrow belt **20** being more narrow than a diameter of the container **12**, ensures that there is adequate space for the closure lift mechanisms **66** to move upwards on the lateral sides of the narrow belt **20**. The closure lift mechanisms **66** may be positioned to locate the guide wheels **68** to match a general position of a bottom ridge **16** of the bottom **14** of the container **12**, such that an annular groove of the guide wheels **68** receives the bottom ridge **16** of the container **12**. FIG. **4C** depicts the bottom ridge **16** of the container **12** being positioned within the annular groove of the guide wheels **68**.

(31) FIGS. **5A-5B** are front, partial cross-sectional and side, partial cross sectional view illustrations, respectively, of a narrow belt **20** and lifting device **64** of the inline container filling machine **10** of FIG. **1**, in accordance with the first exemplary embodiment of the present disclosure. The use of the guide wheels **68** to raise the container **12** off the narrow belt **20** may not only lift the container **12** vertically off the narrow belt **20**, but it may also allow for the container **12** to be rotated when it is in this lifted position. Specifically, when the container **12** is raised towards a seaming device, for example, the seaming device may contact the upper rim of the container **12** and rotate it along the container's **12** vertical axis, thereby allowing the seaming device to seam the lid around the circumference of the container **12**. The guide wheels **68** may continue to support and guide the container **12** as it is rotated, whereby the bottom ridge **16** of the container **12** is moved through the annular groove of each guide wheel **68** used. It is noted that the number of guide wheels **68** used may vary depending on the machine **10**. For example, there may be four wheels, two on each side of the belt for aluminum cans, where the guide wheels **68** match the bottom profile of the cans, so not to damage the cans while spinning them, from the upwards force that is required to seam the cans. Moreover, different containers **12** may have differently-shaped grooved wheels **68**. Additionally, for containers **12** that do not need to be spun or rotated in order to be sealed, such as glass bottles which receive a cap which is ‘crowned’ on the bottle, the grooved wheels **68** may be replaced with a solid plate which pushes the bottle up to crimp the crown onto the bottle.

(32) While the use of the lifting device **64** with closure lift mechanisms **66** and guide wheels **68** (or another container-interfacing structure) is described relative to the use of sealing the container **12**, it is noted that these devices can also be used for any other portion of the container processing, including aspects of processing which requires a container **12** to be lifted off the belt **20** or otherwise moved from a belt. For example, this same technique can be used to raise the containers up to a filling head, below a counter pressure tank, fill the containers, and then drop them back down onto the belt, where the belt would carry them to the closure stations while the next batch of cans or bottles are being filled. The use of these devices can improve the processing speed of the

machine **10**. For example, these devices with the narrow belt **20** design could more than double the throughput in most cases than what is currently on the market, in the same overall space of the machine **10**.

(33) FIG. **6** is a side view illustration of a rotary container filling machine, in accordance with the first exemplary embodiment of the present disclosure. FIG. **7** is an elevated, front isometric view illustration of the rotary container filling machine of FIG. **6**, in accordance with the first exemplary embodiment of the present disclosure. FIG. **8** is an elevated, side isometric view illustration of the rotary container filling machine of FIG. **6**, in accordance with the first exemplary embodiment of the present disclosure. In comparison to FIGS. **1-5B**, which disclose an inline filling machine **10**, FIGS. **6-8** disclose a rotatory filling machine whereby the containers are moved along a rotary path during the fill operation. It is noted that the structures, devices, and functionality of the machine **10** as described relative to the inline filling machine **10** FIGS. **1-5B** can be equally applied to the use of a rotary filling machine as depicted in FIGS. **6-8**, whereby a narrow belt design with a lifting device can be used to raise and lower the containers off the narrow belt as needed during the processing operation.

(34) FIGS. **9-10** are plan view illustrations of a container filling machine **110**, in accordance with a second exemplary embodiment of the present disclosure. The second exemplary embodiment may have numerous similarities to the first exemplary embodiment of FIGS. **1-8**, all of which are considered within the scope of either embodiment. As shown in FIGS. **9-10**, a container filling machine **110**, which may be referred to herein as ‘machine **110**’, includes at least three rotary stations, including a rotary entry station **120**, a rotary filling station **130**, and a rotary gassing station **140**, which are used to fill containers **12** with a fluid, then gas and seal the containers **12**. The rotary entry station **120** receives an empty container **12** which has an open top (no lid). The rotary entry station **120** includes upper and lower plates **122**, **124** which hold the container **12** and move it in a counter-clockwise direction, as indicated by the arrow in FIG. **9**. The empty containers **12** travel around the rotary entry station **120** and to the rotary filling station **130** which holds the containers **12** and fills them with fluid using one or more filling heads **132**, as the containers **12** move around the rotary filling station **130**.

(35) Once the containers **12** are filled, they are transferred to the rotary gassing station **140** which supply a quantity of CO₂ gas to the containers **12** (to prevent oxygen from negatively affecting the fluid within the containers **12**) and place a lid **154** on the containers **12**. The rotary gassing station **140** may be positioned at an immediate exit of the filling station **130**, such that the containers **12** transfer directly from the filling station **130** to the rotary gassing station **140** without traversing on additional lengths of belts, conveyers, or other devices. In one example, the gassing station **140** is rotationally interfacing with the filling station **130** such that as containers **12** move around the filling station **130**, they exit the rotary filling station **130** and are transferred directly to the rotary gassing station **140**. The rotary gassing station includes top and bottom plates **142**, **144** which hold the containers **12** as they move in a counter-clockwise rotational direction, as shown by the arrow in FIG. **9**. During the rotation of the containers **12** around the rotary gassing station **140**, CO₂ gas (or another gas) is supplied through the rotary gassing station **140** itself while a lidding system (not shown) places a lid **154** on the fluid and gas filled container **12**. The container **12** then exits the rotary gassing station **140** and the full containers **12** with lids **154** placed thereon are moved along a belt **112** towards a seaming station where the lids **154** are sealed to the top of the containers **12**.

(36) The details, components, and operation of the rotary gassing station **140** are described in further detail in FIGS. **11-22**.

(37) As shown in FIG. **11** and FIG. **12**, the top and bottom plates **142**, **144** of the rotary gassing station **140** are positioned substantially at a bottom and top of the containers **12**. Each of these plates **142**, **144** may be supported with one or more bars which are connected to one or more lift cylinders **146**, such that the relative vertical position of the top plate **142** and the bottom plate **144** are adjustable. This ability to move the top and bottom plates **142**, **144** relative to one another

allows for the rotary gassing station **140** to be adjustable to differently-sized containers **12**, such as 12 oz aluminum cans versus 16 oz aluminum cans. For example, FIG. **11** illustrates the rotary gassing station **140** in use with 16 oz containers whereas FIG. **12** illustrates the rotary gassing station **140** in use with 12 oz containers **12**, which have a smaller height. The lift cylinders **146** may allow for movement of the top plate **142** while the bottom plate **144** remains substantially vertically stationary.

(38) As can also be seen in the figures, the top plate **142** and the bottom plate **144** may have star wheel pockets which allow for the top and/or bottom of the container **12** to be exposed or accessible. In particular, the bottom plate **144** may have pockets which allow the container **12** to be raised in a linear motion upwards to engage with the lid that is placed on top of the top star wheel **142**. When the bottom of the container **12** is moved rotatably by the rotary gassing station **140**, it comes in contact with a raising ramp (discussed later) which causes the container **12** to move vertically upwards. This action allows the container **12** to be lidded, as discussed further herein.

(39) FIG. **13** illustrates the rotary gassing station **140** in use with a lidding system **150**, which supplies a quantity of lids **154** through a chute **152**, such that the lids **154** are dispensed on to the top plate **142** of the rotary gassing station **140**.

(40) FIGS. **14-15** illustrate the gassing structures of the rotary gassing station **140**. As shown, the top plate **142** includes a central gas inlet **160** which receives the CO₂ gas (or another gas) and directs the gas to flow into a manifold **162**. The manifold **162** transfers the gas to the degassing pockets **164** of the top plate **142**, in which a container **12** may be positioned. One or more fins or flow structures may be positioned within the manifold **162** to assist with preventing cross flow or other gas movement which is undesirable. As a constant or near-constant supply of gas is provided through the manifold **162**, the exposed tops of the containers **12** are gassed which prevents ambient air or oxygen from gaining access to the containers **12**. The goal is to replace atmosphere from the “head space”, the area from the top of the fluid in the container, to the bottom of the lid **154**, and replace it with CO₂ gas before the lid **154** is placed on top of the container **12**. While this occurs, the lid **154** may be positioned on the top plate **142**, and the container **12** is moved upwards to make contact with the lid **154** and remove it from the top plate **142**, with the lid **154** positioned over the opening of the container **12**. Even when the lid **154** is not sealed, its position on the opening of the container **12** may act as a physical barrier to prevent gas flow into or out of the container **12**. FIGS. **16A-16C** illustrate additional images of the degassing process, where the top plate **142** holds down the lids **154** that are placed into the top plate **142** pockets. As the gas is moved through the manifold, a side shield (FIG. **16A**) keeps the gas confined at the degassing pockets **164**.

(41) FIGS. **17A-17C** illustrate the movement of the container **12** to capture the lid **154** during the degassing process. As shown in FIG. **17A**, as the gas is flowed into the degassing pockets **164**, the headspace of the container **12** is filled with the gas. Initially, the container **12** is positioned in a lowered position (FIG. **17A**) such that the headspace is open to receiving the gas. As the container **12** rotates around the rotary gassing system **140**, the container **12** is moved up a ramp **170** (FIG. **17C**) which moves the top of the container **12** vertically. This position is shown in FIG. **17B**. In one example, the container **12** will raise up approx. ½ inch within 1 second to meet the lid **154**. During this process, the upper plate **142** and the lower plate **144** remain substantially fixed in vertical height (but the upper plate **142** is adjustable for different container sizes) but the containers **12** are moveable within the pockets of the plates **142**, **144** to allow for contact with the ramp to move the container **12** upwards. FIG. **17C** illustrates the movement of the container **12** on the ramp **170**, whereby as the container **12** is rotated with the top and bottom plates **142**, **144**, the bottom of the container **12** contacts an inclined ramp **170** and is moved vertically upwards (as indicated by the arrows in FIG. **17C**). This movement of the container **12** on the ramp **170** decreases the headspace of the container **12** and causes the top of the container **12** to contact the lid. The ramp **170** may be positioned to raise the container **12** from the plane of the discharge from the rotary filling system **130** to the plane of the belt **112** (FIG. **9**), such that the transition closes the gap between the top of

the container **12** and the bottom of the lid **154** to trap the gas within the container **12** headspace. Once on the belt **112**, the container **12** with lid **154** may be moved to a closing or seaming station, which seals the lids **154** to the containers **12**.

(42) FIGS. **18A-18D** are illustrations of the progression of the container **12** through the rotary gassing system **140**. As can be seen, the rotary gassing system **140** moves counter clockwise and the containers **12** enter towards the left. As the container **12** moves around the rotary filling system **140** it is gassed. When the container **12** reaches the right-hand side of the rotary filling system **140**, it is evacuated from the rotary filling system **140** by removing the container **12** with lid from the wheel. The container **12** then moves along a belt to the seaming station, or another station. The star wheels of the rotary gassing station **140** may move at various speeds. In one example, they move at 20 RPM. The containers **12** enter the wheel with no lid, and the fluid is positioned about $\frac{3}{8}$ " from the top of the container **12**. The containers **12** then exit the wheel with a lid on top at approximately 50 containers per minute.

(43) During the rotation of the containers **12** and the vertical movement by means of the ramp, the containers **12** effectively pick up the lid from the top wheel **142**. FIG. **19** illustrates a progression of the container **12** picking up the lid from the top wheel **142** and being evacuated from the rotary gassing station **140**.

(44) FIGS. **20A-20B** illustrate how the lid **154** is released from the top wheel **142** to the container **12**. As shown, a stack of lids **154** are held within a chute **152** of the lidding system **150**. As the lids **154** are moved down towards the container (not shown) by gravity, a lid stop feeding gate **180** stops the feed of the stack of lids **154** and allows one lid **154** to pass below. The lid stop feeding gate **180** may move horizontally relative to the movement of the lids **154** which allow it to separate the stacked lids **154**. The lid stop feeding gate **180** has a lid separator knife shelf **182** which contacts the lids **154** to separate them, allowing the one lid **154** to drop below through a lid delivery shelf **184** and to rest on a lid resting shelf **186**. At this point, when a container (not shown) is positioned within the pocket of the top plate **142** of the rotary gassing system **140**, the single lid **154** is held on the lid resting shelf **186** until the container is moved vertically upwards by the ramp (FIG. **19**). The top edge of the container makes contact with the lid **154** to move it off the lid resting shelf **186**, at which point the container with lid **154** are evacuated from the rotary gassing system **140**, as shown by the progression in FIG. **19**.

(45) FIG. **20A** illustrates the lid stop feeding gate **180** in an open position, whereby the stack of lids **154** can move vertically downwards, whereas FIG. **20B** illustrates the lid stop feeding gate **180** in a closed position, preventing the stack of lids **154** from moving downwards (while allowing a single lid **154** to move downwards). FIGS. **20C-20D** illustrate another view of the lid stop feeding gate **180**, where the mechanical aspects of the device can be seen. FIG. **20C** illustrates the lid stop feeding gate **180** in an open position, where the lid stop feeding gate **180** is horizontally removed from the stack of lids **154** while FIG. **20D** illustrates the lid stop feeding gate **180** in a closed position, where it is moved horizontally inwards towards the stack of lids **154** and lid separator knife shelf **182** separates one lid (not shown) from the remaining stack of lids **154**.

(46) FIGS. **21A-21L** illustrate additional images of the progression of the lid **154** being evacuated from the rotary gassing system **140** with a container (not shown for clarity). As can be seen, the lid **154** is moved along an outer guide **190** and an inner guide **192** of the top plate **142** towards a pocket **148** formed through the top plate **142**. The lid **154** is held by the edges of the lid **154** contacting the outer and inner guides **190**, **192**. When the lid **154** is moved and approaches the pocket **148**, the edges of the lid **154** contact an inverted ramp **194** which pushes the lid **154** vertically downwards. As the top plate **142** rotates, the lid **154** is pushed further into the pocket **148** and the inverted ramp **194** biases the lid **154** downwards until the top edge of the lid **154** is positioned substantially below the bottom edge of the outer guide **190**, as shown in FIG. **21F**. At this point, the lid **154** is evacuated from the top plate **142** in a substantial radial direction, where the lid **154** passes under the outer guide **190**, as shown in FIGS. **21G-21H**. FIGS. **21I-21L** depict the

same progression but with a stack of lids **154**.

(47) While some of the figures of this embodiment, including FIGS. **9-10**, depict the rotary gassing system being used with a rotary filling system, it is noted that the rotary gassing system can also be used with an in-line filling system. For example, FIG. **22** depicts the rotary gassing system used with an inline filler having a counter pressure filling tank **200** which fills containers **12**. The containers **12** are moved from the inline filler tank **200** to the rotary gassing system **240** which operates in the same manner as previously described. When the containers **12** are evacuated from the rotary gassing system **240**, they are moved along a belt to the next operation, such as a seaming station which seals the lids to the containers **12**.

(48) FIG. **22** further shows the use of a star wheel **250** which can be used to manage the flow of containers **12** through the system. The star wheel **250** is positioned along the path of movement of the containers **12** and proximate to various stations. For example, in FIG. **22**, the star wheel **250** may be positioned at a seaming station, where as the container **12** progresses on the belt, the container **12** enters one pocket of the star wheel **250** which captures the container **12** and prevents it from bouncing backwards out of position of the seaming head (shown in FIGS. **3A-3B**). After the seaming head completes the seam of the lid to the container **12**, as described relative to FIGS. **3A-3B**, the next container **12** enters the next pocket of the star wheel **250** which helps move the recently seamed container **12** from the seaming head. Without the star wheel **250**, when the recently seamed container **12** lowers on to the belt, it may sit there for a longer period of time than desired, i.e., a period of time until it gets traction on the belt to move out of the way. While the use of the stop gate (reference character **40** in FIGS. **3A-3B**) can be used to control movement of the containers **12**, the containers **12** may be susceptible to bouncing, moving slightly backwards, or other imprecise movements when they contact the stop gate. Thus, the use of the star wheel **250** helps increase the throughput of the containers **12** through the seaming station by controlling the movement of the containers **12** better and more precisely. The star wheel **250** may also be used in other locations along the processing system to provide more precise container **12** movement control.

(49) In a third exemplary embodiment of the present disclosure, an apparatus for closing fluid containers is provided to overcome the shortcomings identified in the Background of the present disclosure. FIG. **23A** is a top view illustration of an apparatus for closing fluid containers, in accordance with a third exemplary embodiment of the present disclosure, and FIG. **23B** is a top view diagram of the apparatus for closing fluid containers of FIG. **23A**, in accordance with the third exemplary embodiment of the present disclosure. In particular, FIG. **23B** illustrates a simplified operation of the apparatus for closing fluid containers **300**, which may be referred to herein as apparatus **300**. As shown in FIGS. **23A-23B**, a container infeed **310** is used to receive a plurality of empty fluid containers **302** in the apparatus **300**. The empty fluid containers **302** are delivered from the container infeed **310** to a filling station **320** using a transfer star wheel **312**, where the empty fluid containers **302** are diverted from the substantially linear path of the container infeed **310** to a rotational path of the filling station **320**. Within the filling station, the empty fluid container **302** is at least partially filled with a quantity of fluid in an interior space of the fluid container **302**, which results in the empty fluid container **302** becoming a fluid filled container **304**.

(50) The fluid filled container **304** then passes to a rotary gassing station **330** which receives the fluid filled container **304** from the filling station **320**. In the rotary gassing station **330**, the fluid filled container **304** is subjected to a degassing operation, where a quantity of gas, such as CO₂, is applied to the opening of the fluid filled container **304** to replace ambient air which is proximate to the fluid within the top space of the fluid filled container **304**, such that the interior space of the fluid filled container **304** contains fluid and gas therein, to thereby provide a fluid and gas filled container **306**, as shown in FIG. **23B**. A lid from a lid chute **332** is then provided to the fluid and gas filled container **306** within the rotary gassing station **330** in a fast manner to close the opening of the fluid and gas filled container **306** before the CO₂ gas or other gas escapes.

(51) After the fluid and gas filled container **306** has a lid positioned on an opening thereof, it is then transferred directly to a rotary seaming station **340** which is positioned proximate to the rotary gassing station **330**. The rotary seaming station **340**, in general operation, receives the fluid and gas filled container **306** and seams the lid to the container **306**, to thereby produce a closed container **308**. When the lid is seamed to the opening of the container **306**, the fluid and the gas in the interior space of the container **306** is sealed from the outside atmosphere. Seaming the lid to the fluid and gas filled container **306** may include the use of a seaming chuck and roller wheels, as described in greater detail relative to FIGS. **24-26C** and **28A-28B**. While the subject disclosure discloses the use of a seaming station to seam a lid to a container, it is noted that the rotary seaming station **340** may also be capable of closing containers by other processes and techniques, such as with caps, threadable lids, or other structures.

(52) The lid seaming operation occurs in the rotary seaming station **340**, but the fluid and gas filled container **306** is initially placed in contact with the rotary seaming station **340** while it is still located in the rotary gassing station **330**. In particular, while the fluid and gas filled container **306** is positioned in a recess of a star wheel of the rotary gassing station **330**, at least one container piston of the rotary seaming station **340** contacts the fluid and gas filled container **306** and moves it into the radial path of the rotary seaming station **340**, where the fluid and gas filled container **306** has the lid seamed. After the lid is sealed to the opening, the closed container **308** may then exit the rotary path of the rotary seaming station **340** on a container discharge path **350**.

(53) In many instances, it is highly desirable to place the lid on the opening of the fluid and gas filled container **306** at the earliest opportunity possible, and then seam the container **306** without delay. As previously described relative to other figures, the apparatus **300** may utilize a rotary gassing station where the lid is dropped through a top plate and on to the opening of the container **306**. To ensure that the lid is sealed at the earliest time possible, the container **306** is to be transferred to the rotary sealing station **340** at the quickest time possible. To achieve this fast timing of the handoff between the gassing and the seaming stations, as shown in FIG. **23B**, the pitch diameters of the rotary gassing station **330** and the rotary seaming station **340** may be coincidence, in that, the footprints of the rotary structures forming the rotary gassing station **330** and the rotary seaming station **340** overlap. In this overlapping position, where the rotary seaming station **340** is at a direct output of the rotary gassing station **330**, an unsealed but lidded container **307** may be transferred from the rotary gassing station **330** to the rotary seaming station **340**.

(54) FIGS. **24-26C** further describe the handoff process of the unsealed but lidded container **307**. With reference to FIGS. **23A-26C**, the rotary seaming station **340** has at least one container piston **342** which is positioned on a carriage **344** which rotates around a central vertical axis of the rotary seaming station **340**. The lidded but unsealed container **307** is positionable on the top of the container piston **342**, as shown in FIG. **24**. It is noted that the rotary seaming station **340** may include any number of container pistons **342**, such as three container pistons **342** as shown in the figures. The container piston **342** is positioned vertically below at least one seaming chuck **346** and a plurality of seaming wheels **348**. The seaming chuck **346** and the seaming wheels **348** are used to removably contact the container **307** in order to seam the lid to the opening of the container **307**. As shown in FIGS. **24-25**, in the exemplary instance of the apparatus **300**, the rotary seaming station **340** may have at least three container pistons **342** and at least six seaming wheels **348**, where the container **307** is moved to contact at least two of the at least six seaming wheels **348** and the seaming chuck **346** simultaneously when the lid is seamed to the opening of the container **307**.

(55) Each of the container pistons **342** are movable fully around a center axis of the rotary seaming station **340**, such that they can rotate without limitation around the center axis of the rotary seaming station **340** in a repeatable manner to provide continuous seaming of the containers **307**. It is noted that it may be preferable for the container pistons **342** to be container lift pistons **342** which raise and lower in a vertical direction, such that the container **307** can be raised or lowered vertically during the seaming operation. In another example, it may be possible to maintain the container

pistons **342** as vertically stationary structures where other components of the rotary seaming station **340** move vertically. It is noted that the turret of the various rotary stations may be adjustable to account for containers **307** having different heights and sizes.

(56) With reference to FIGS. **23A-23B** and **26A-26C**, the handoff operation of the container **307** is described. The rotary gassing station **330** may include at least one star wheel with a plurality of container receiving areas, but in most situations, the rotary gassing station **330** will include an upper star wheel **334** and a lower star wheel **336**, where each of the star wheels **334**, **336** have a container receiving area, e.g., a portion of the star wheel **334**, **336** which is contoured or has a cut-out portion which is sized to fit the container **307**, such that when the star wheels **334**, **336** are rotated, the container **307** is in contact with the container receiving area and moves in the rotational path.

(57) The movement of the container **307** by the container piston **342** of the rotary seaming station can be understood as being between at least two radial positions: a first radial position where the container **307** is positioned within one of the plurality of container receiving areas of the star wheels **334**, **336**, and thus still within the rotary gassing station, and a second radial position where the lid is seamed to the opening of the container **307**. The container piston **342** may also move the container **307** to additional positions, such as a third radial position where a guide star wheel **352** exits the container **307** from the rotary seaming station **340** to the container discharge path **350**.

(58) The container piston **342** first contacts the container **307** while it is in the rotary gassing station **330** from a position underneath the container **307**. While the container **307** is still within the rotary gassing station **330**, the container piston **342** maintains contact with the container **307** and moves it away from the rotary gassing station **330**, e.g., out of the rotational path of the rotary gassing station **330**, and on to the rotary path of the rotary seaming station **340**. The container piston **342** maintains contact with the container **307** while it is moved around the rotary seaming station **340**, such as during a raising and a lowering vertically of the container **307** to the seaming chuck **346** and seaming wheels **348**. As such, the container piston **342** effectively holds the container **307** between the two aforementioned radial positions, including the first radial position when the container **307** is within the rotary gassing station **330** to the second radial position when the lid is seamed to the opening of the container **307** at or near the conclusion of the seaming process. In the first radial position, the container **307** may be in contact with at least one of the plurality of container receiving areas of the upper and lower star wheels **334**, **336** while it is positioned in contact with the container piston **342** of the rotary seaming station **340**.

(59) Unlike conventional seaming apparatuses which utilize a belt or similar conveyance mechanism to transport containers from a gassing station to a seaming station, where the conventional seaming station is positioned a distance away from the gassing station, the apparatus **300** does not use a belt or intermediary conveyance mechanism. In particular, the apparatus **300** utilizes rotary gassing and seaming stations **330**, **340** which are in direct working operation with one another, since the rotary gassing and seaming stations **330**, **340** have a pitch diameter which is coincident with one another, and the container piston **342** of the rotary seaming station **340** can move the container **307** from the rotary gassing station **330** directly to the rotary seaming station **340** without a belt or conveyance device. Specifically, as described herein, the use of the container piston **342** to transport the container **307** from the pockets or recesses of the star wheels **334**, **336** of the rotary gassing system **330** to the rotary seaming station **340** means that a traditional conveyance belt which is used with conventional devices is neither needed nor used.

(60) A sequential depiction of the process is shown in FIGS. **27A-27F**, which are top view illustrations of the rotary seaming station of the apparatus for closing fluid containers **300**, in accordance with the third exemplary embodiment of the present disclosure. In particular, FIGS. **27A-27F** depict the processing of an exemplary container **309**. As shown in FIG. **27A**, the container **309** is positioned within the rotary gassing station **330**, as it has already been filled with fluid in the rotary filling station **320**. The container **309** has also been supplied with a lid positioned

over the top opening of the container 309, but the lid has not yet been sealed or seamed to the container 309. In FIG. 27B, the container 309 has been moved into the position where it is contacted by a container piston 342 (not shown in FIG. 27B, but shown in FIGS. 24-26C). In this location, the container 309 is positioned in the coincidence position of the pitch diameters of the rotary gassing and seaming stations 330, 340, such that it is contacted by the star wheels of the rotary gassing station 330 and the container piston of the rotary seaming station 340. In FIG. 27C, the container 309 has been moved by the container piston 342 from the rotary gassing station 330 and to the rotary seaming station 340, where it is seamed to seal the fluid and gas inside the container. In FIG. 27D, the container 309 has moved to an exit location of the rotary seaming station 340, where it contacts a guide star wheel 352 of the discharge path 350. In FIG. 27E, the container 309 has started moving on the discharge path 350, and in FIG. 27F, the container 309 is moving down the discharge path 350 to the next operation, such as cleaning or packaging.

(61) FIGS. 28A-28B are plan view illustrations of the rotary seaming station of the apparatus for closing fluid containers 300, in accordance with the third exemplary embodiment of the present disclosure. In particular, FIGS. 28A-28B illustrate the container pistons 342 being capable of moving in a vertical direction, such that the container 307 can be positioned to contact the seaming chuck 346 and the seaming wheels 348 of the rotary seaming station 340. As shown, in FIG. 28A the container 307 on the container piston 342 moves vertically upwards to position the top of the container 307 proximate to the seaming chuck 346 and seaming wheels 348. In this position, the seaming wheels 348 may move inwards towards each other to contact the container 307 and seam the lid to the container 307 sidewall. When the operation is complete, the seaming wheels 348 may move apart, as shown in FIG. 28B, and the container piston 342 moves vertically downwards towards the carriage 344. The container 307 may then be offloaded from the rotary seaming station 340 to the discharge path.

(62) It should be emphasized that the above-described embodiments of the present disclosure, particularly, any “preferred” embodiments, are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the disclosure. Many variations and modifications may be made to the above-described embodiment(s) of the disclosure without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and the present disclosure and protected by the following claims.

Claims

1. An apparatus for closing fluid containers, the apparatus comprising: a rotary gassing station, wherein in the rotary gassing station, at least one container containing a quantity of fluid within an interior space thereof receives a quantity of gas in the interior space; a rotary seaming station positioned proximate to the rotary gassing station, wherein in the rotary seaming station, a lid is seamed to an opening of the at least one container to seal the quantity of fluid and the quantity of gas in the interior space; and at least one container piston of the rotary seaming station movable around a center axis of the rotary seaming station, the at least one container piston holding the at least one container between a first radial position when the at least one container is within the rotary gassing station to at least a second radial position when the lid is seamed to the opening of the at least one container.
2. The apparatus of claim 1, wherein the rotary gassing station further comprises a star wheel with a plurality of container receiving areas, wherein in the first radial position, the at least one container is positioned within one of the plurality of container receiving areas.
3. The apparatus of claim 2, wherein the star wheel of the rotary gassing station further comprises upper and lower star wheels, each with the plurality of container receiving areas, wherein the at least one container is in contact with at least one of the plurality of container receiving areas of the

upper and lower star wheels while it is positioned in contact with the at least one container piston of the rotary seaming station.

4. The apparatus of claim 1, wherein the at least one container piston further comprises a container lift piston, wherein the container lift piston raises and lowers in a vertical direction.

5. The apparatus of claim 1, wherein the rotary seaming station further comprises a seaming chuck and at least one seaming wheel removably in contact with the at least one container when the lid is seamed to the opening of the at least one container.

6. The apparatus of claim 5, wherein the rotary seaming station further comprises at least three container pistons and at least six seaming wheels, wherein the at least one container contacts at least two of the at least six seaming wheels and the seaming chuck simultaneously when the lid is seamed to the opening of the at least one container.

7. The apparatus of claim 1, wherein a pitch diameter of the rotary gassing station coincides with a pitch diameter of the rotary seaming station.

8. An apparatus for closing fluid containers, the apparatus comprising: a container infeed; a filling station receiving at least one container from the container infeed, wherein in the filling station, the at least one container is at least partially filled with a quantity of fluid in an interior space of the at least one container; a rotary gassing station receiving the at least one container from the filling station, wherein in the rotary gassing station, the at least one container containing the quantity of fluid within the interior space thereof receives a quantity of gas in the interior space; a rotary seaming station positioned proximate to the rotary gassing station, wherein in the rotary seaming station, a lid is seamed to an opening of the at least one container to seal the quantity of fluid and the quantity of gas in the interior space; and at least one container piston of the rotary seaming station movable fully around a center axis of the rotary seaming station, the at least one container piston holding the at least one container between at least first radial position when the at least one container is within the rotary gassing station to at least a second radial position when the lid is seamed to the opening of the at least one container.

9. The apparatus of claim 8, wherein the rotary gassing station further comprises a star wheel with a plurality of container receiving areas, wherein in the first radial position, the at least one container is positioned within one of the plurality of container receiving areas.

10. The apparatus of claim 9, wherein the star wheel of the rotary gassing station further comprises upper and lower star wheels, each with the plurality of container receiving areas, wherein the at least one container is in contact with at least one of the plurality of container receiving areas of the upper and lower star wheels while it is positioned in contact with the at least one container piston of the rotary seaming station.

11. The apparatus of claim 8, wherein the at least one container piston further comprises a container lift piston, wherein the container lift piston raises and lowers in a vertical direction.

12. The apparatus of claim 8, wherein the rotary seaming station further comprises a seaming chuck and at least one seaming wheel removably in contact with the at least one container when the lid is seamed to the opening of the at least one container.

13. The apparatus of claim 12, wherein the rotary seaming station further comprises at least three container pistons and at least six seaming wheels, wherein the at least one container contacts at least two of the at least six seaming wheels and the seaming chuck simultaneously when the lid is seamed to the opening of the at least one container.

14. The apparatus of claim 8, wherein a pitch diameter of the rotary gassing station coincides with a pitch diameter of the rotary seaming station.

15. An apparatus for closing fluid containers, the apparatus comprising: a rotary seaming station; and at least one container lift piston in the rotary seaming station, wherein the at least one container lift piston holds at least one container at and between a first radial position when the at least one container is within a rotary gassing station to at least a second radial position when a lid is seamed to an opening of the at least one container in the rotary seaming station.

16. The apparatus of claim 15, wherein the rotary gassing station further comprises a star wheel with a plurality of container receiving areas, wherein in the first radial position, the at least one container is positioned within one of the plurality of container receiving areas.

17. The apparatus of claim 16, wherein the star wheel of the rotary gassing station further comprises upper and lower star wheels, each with the plurality of container receiving areas, wherein the at least one container is in contact with at least one of the plurality of container receiving areas of the upper and lower star wheels while it is positioned in contact with the at least one container piston of the rotary seaming station.

18. The apparatus of claim 15, wherein the at least one container lift piston raises and lowers in a vertical direction to move the at least one container vertically.

19. The apparatus of claim 15, wherein the rotary seaming station further comprises a seaming chuck and at least one seaming wheel removably in contact with the at least one container when the lid is seamed to the opening of the at least one container.

20. The apparatus of claim 15, wherein a pitch diameter of the rotary gassing station coincides with a pitch diameter of the rotary seaming station.
